

Evolutionary Stress Tests Reveal Parent-Locked Tonic Modulation in Neural Cellular Automata

Anonymous Author(s)
anonymous@example.com

Abstract

We set out to test whether closed-loop control of global chemical signals makes regenerative Neural Cellular Automata (NCAs) more robust, and built the evaluation to prove it: recurring multi-block lesions larger than the perception radius, a small evolved controller for three broadcast modulator channels, held-out damage seeds. On a single trained model the experiment looked like a success. Crossing five independently trained parent seeds (two independent controller evolutions each; preregistered) dissolves that success into three findings. First, the robustness we had attributed to control is mostly a property of training: parents trained with channels present beat unmodulated siblings in five of five seeds (median final-Hamming reduction 0.008, up to ≈ 0.14 on the most fragile parent) with modulation pinned to neutral. Second, evolved controllers add little on top: on four parents the effect is absent or noise-level; on the fragile parent both evolutions reproducibly help (0.035–0.038 $\rightarrow$ 0.029–0.030, survival 0.975 $\rightarrow$ 1.00) — yet its tonic vector is nearly identical (cosine 0.99) to a sibling’s that does not help, so the benefit is an interaction with parent dynamics, not a distinctive operating point. All ten controllers emit flat, nonzero, parent-specific constants with no lesion-locked response. Third, these constants are organism-locked: a single-donor probe penalizes five of five siblings (lethally in two), a full 5×5 transfer matrix shows most transfers harmful and none better than the recipient’s own controller, and injecting each donor’s tonic constant directly — no controller at all — reproduces its transfer outcomes in 46 of 50 cells, including every lethal one. Single-parent evaluation would have approved all of it. The attribution protocol itself is the transferable artifact.

1 Introduction

Growing Neural Cellular Automata (GNCAs) grow a target morphology from a single seed and regenerate it after damage using one local, differentiable update rule [1]. The catch is perception: a cell sees only its immediate neighborhood, so wounds larger than a few cells contain tissue with no signal to integrate, and a severed fragment has no way to know what it was part of. The natural remedy is a global signal. Prior work, our own single-parent study included (Appendix H), adds global modulator channels and evolves release policies for them — and reports the outcome on one trained model.

That single-model habit is the problem this paper is about. When a modulated NCA outperforms an unmodulated one, three effects are indistinguishable: the channels being *present during training*, the controller *acting at run time*, and whatever *transfers* to another organism. Our single-parent study exhibited the failure firsthand: it credited closed-loop control for a 2× improvement that a later pilot traced elsewhere — the controller transferred lethally to siblings while zero-modulation channel parents beat the unmodulated baseline in every seed (Appendix C). The improvement was real; the attribution was wrong.

We therefore ran the attribution study that the original design needed: preregistered, with the primary statistic and decision thresholds fixed before data collection. Five parent seeds, each trained from scratch as a $K=0$ pair and a channel-aware $K=3$ pair; *two* independent controller evolutions per channel parent; four conditions per seed, all on held-out damage seeds; then, post-hoc, a full 5×5 transfer matrix and a tonic-transplant condition that replaces each controller with its realized mean output (Section 3). Preregistrations and analysis scripts are in the repository history, timestamped before the runs.

Contributions.

1. A preregistered robustness-attribution protocol for regenerative NCAs: adversarial damage calibrated to exceed perception, crossed with parent-seed variation, decomposed into E_{train} , E_{ctrl} , E_{transfer} , with calibration probes that keep the evolution landscape interpretable (Appendix F).
2. The attribution, settled causally: training with channels is the robust cause (5/5 seeds); the controller effect is parent- dependent (absent or noise-level on four parents, reproducible on the fragile one); the evolved artifact is a parent-specific tonic constant that penalizes every sibling it meets (lethally in two), and injecting the constant alone reproduces its transfer outcomes — the matrix (23/40 harmful, none beating the recipient’s own) closes the case.

2 Related Work

Neural cellular automata. GNCAAs grow patterns from a seed and regenerate after damage [1], with extensions to self-classification [2] and textures [3]. All rely on local perception, and regeneration degrades beyond the perception radius — the regime our benchmark exploits. Signal channels [4], goal conditioning [5], and information-dynamical analyses of self-maintenance [6] all treat the signal as fixed input or emergent byproduct, never a controlled output; we ask what is actually helping when such a channel *appears* to help.

Search over evaluation regimes. Our design follows a concern shared by novelty search [7] and quality-diversity optimization [8]: a fixed objective can misdirect search and hide failure modes. Environment-generation methods co-evolve challenges with solutions — POET [9], PAIRED [10], ACCEL [11]; our benchmark does not yet evolve the damage distribution and serves as the stationary baseline such co-evolution can be judged against.

3 Experimental design

Damage regime. Recurring multi-block lesions — every 150 steps, $n=4$ contiguous 16×16 blocks at seeded positions, $T=2000$ — exceed the perception radius, so wound interiors have no living neighbors (Appendix E). Damage seeds 0–7 drive evolution; held-out seeds 10000–10007 drive all reported numbers; the schedule is deliberately adversarial (Appendix F).

Parents and controllers. Per seed s we train a $K=0$ and a channel-aware $K=3$ parent from scratch (Appendix D); the $K=3$ parent’s three global modulator channels are the only non-local pathway, each carrying a *tonic* (slow baseline) and *phasic* (event-triggered, fast-decaying) component — the distinction from biological neuromodulation. A 259-parameter controller ($4 \rightarrow 32 \rightarrow 3$, tanh) reads four target-free grid statistics and sets release every 10 steps; each parent gets *two* independently evolved controllers via CMA-ES [12] in Evosax [13] (population 64, $\sigma_0=0.01$, 300 generations, different evolution seeds, event-weighted Hamming objective; Appendix G).

Preregistered comparison. All conditions run on the same held-out damage seeds (5 condition seeds $\times$ 8 damage seeds) from a shared $t=0$ state grown by the $K=3$ parent. The primary statistic is $\Delta_s = H(K=3, m=0, s) - H(K=3, \text{own ctrl}, s)$, positive when the evolved controller helps *its own* parent. Fixed before data collection: *controller effect supported* if $\Delta_s > 0$ in $\geq 4/5$ seeds with sign-consistent differences; *channel-training effect supported* if $H(K=0, s) - H(m=0, s) > 0$ in $\geq 4/5$; *transfer failure supported* if the July controller underperforms the own-controller substantially in $\geq 4/5$. Effects are reported per seed (median and range); no significance claims at five seeds. The defense study reported here fixed the two-evolutions-per-parent rule in its preregistration before launch (experiment_results/20260818_evoseed.defense/PREREGISTRATION.md in the repository history). Controller-output (m_t) series distinguish tonic calibration from event-locked policy (Appendix B). A post-hoc follow-up evaluates the full 5×5 transfer matrix among the defense

parents (both controller replicas, 8 damage $\times$ 3 condition seeds per cell; Table A3) plus a *tonic transplant* injecting each donor’s realized mean m_t as a constant.

4 Results

Table 1 reports the preregistered decomposition; the full per-seed, per-condition numbers behind it are in Appendix A.

Table 1: Final Hamming (mean over 5 condition seeds $\times$ 8 held-out damage seeds). Survival in parentheses where it departs from 1.00. Training helps in 5/5 seeds; the controller effect is parent-dependent.

Parent seed	$K=0$	$K=3, m=0$	Own ctrl (2 runs)	Interpretation
0	0.035–0.036	0.029	0.029–0.030	channel effect; no control effect
1	0.163–0.175 (surv. 0.05–0.08)	0.035–0.038 (surv. 0.975)	0.029–0.030 (surv. 1.00)	channel effect + replicated tonic benefit
2	0.030	0.018–0.021	0.022–0.026	channel effect; controller slightly worse
3	0.049–0.052	0.028–0.029	0.032–0.033	channel effect; controller slightly worse
4	0.042–0.045	0.033	0.034–0.035	channel effect; no control effect

Outcome. Applying the preregistered rule: *channel-training supported* (positive in 5/5 seeds; bar $\geq 4/5$); *transfer failure supported* (penalty 5/5, lethal in 2). The controller effect is classified as *parent-dependent* — preregistered **Outcome C**: absent or noise-level on four parents, reproducibly beneficial on the fragile one (both evolutions: 0.035–0.038 $\rightarrow$ 0.029–0.030, survival 0.975 $\rightarrow$ 1.00).

What the controller actually emits. All ten evolved controllers emit a *constant at a nonzero level*: within-rollout per-channel std of m_t is 0.0002–0.0046 — tonic, with no lesion-locked response, in 10/10 runs — while channel means sit at parent-distinct offsets (Figure A1). Evolution neither collapses to neutral output nor discovers a policy: it finds a parent-specific tonic calibration. That calibration alone does not predict benefit: seeds 1 and 4 emit nearly identical vectors (cosine 0.993), yet only seed 1 benefits.

Cross-parent transfer matrix. The full 5 $\times$ 5 matrix among the defense parents — both replicas, three condition seeds per cell, 40 off-diagonal cells (Table A3, Figure 1) — 23/40 transfers are harmful (survival < 0.9 or final Hamming worse than the recipient’s zero-output baseline by > 0.005), 15 are indistinguishable from zero-output, and 2 beat it. No foreign controller beats the recipient’s own controller beyond noise. Lethal transfer replicates across controller replicas and condition seeds: donor s3 kills recipient s2 in both replicas (survival 0.00); donor s0 on recipient s1 is lethal in e1 and near-lethal in e2. Classifications are stable across condition seeds (Figure 1A).

Tonic alignment structures transfer. All lethal instances pair strongly negative tonic-vector cosines (s3 $\rightarrow$ s2: -0.61 ; s0 $\rightarrow$ s1: -0.93), while the only beneficial transfers — beating zero-output and matching the own controller — occur exclusively within the tonic-aligned pair s4 $\rightarrow$ s1 (cosine $+0.99$, both replicas; adjacent to the diagonal in Figure 1). Across all 40 cells, tonic cosine correlates with transfer penalty ($r = -0.30$) and survival ($r = +0.34$); alignment is not sufficient — some benign cells carry cosines to -0.95 .

Tonic transplant. Injecting each donor’s realized mean m_t as a constant in place of its controller (both replicas, three condition seeds) reproduces the full controller’s outcome in 36/40 off-diagonal cells, including every lethal transfer (the four mismatches are magnitude differences inside already-lethal cells), and matches the recipient’s own controller in all 10 diagonal cases (46/50 overall;

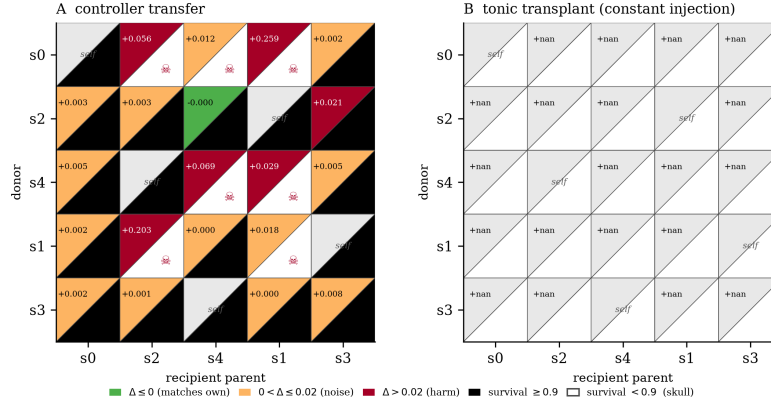


Figure 1: Transfer redesigned for attribution (replica-averaged; per-cell Δ from the recipient’s own controller; survival as the split lower triangle; rows/columns cosine-ordered so the aligned pair s4–s1 borders the diagonal and the anti-aligned pair s3–s2 sits at opposite ends). Panel A: controllers. Panel B: the same matrix with each donor’s tonic constant injected instead — matching A in 46/50 cells, every lethal one included. Values: Table A3.

Appendix A). The transferable component is thus the tonic setpoint; the controller’s small dynamic residual did not measurably contribute, on-parent or off.

5 Discussion

Which component causes robustness. Channel-aware training, not run-time control. Parents that grew up with global channels beat their unmodulated siblings on every seed; the channels are a developmental scaffold, making the organism more robust even when they carry no signal at run time. What evolution adds is narrow: nothing measurable on four parents; on the fragile parent, a final stabilization that both independent evolutions find — and since that parent’s tonic vector is nearly identical to a sibling’s that gains nothing from it, the benefit is not a property of the tonic. We conjecture the fragile parent’s dynamics sit near a stability boundary, so a small tonic push re-stabilizes the attractor (unverified). The broader lesson: held-out *damage* seeds approved every condition above; crossing *parent* seeds is what separated training from control, benefit from harm, and policy from constant.

Why cross-parent transfer fails — and when it does not. The full matrix (Table A3) sharpens the single-donor probe: no foreign controller beats the recipient’s own controller, most transfers are harmful (23/40), and the only beneficial ones stay within the tonic-aligned pair (s4→s1). The tonic transplant makes it causal: what transfers, for good or ill, *is* the constant (36/40 off-diagonal, 10/10 diagonal, every lethal cell included). Each controller’s output is a calibration against its own parent’s channel weights — the same release level drives different dynamics in a sibling; evolution found an operating point for one organism, not a modulation law.

Limitations and future work. Five parents, two evolutions each; one morphology; one damage family; the matrix covers only these parents and replicas; with one controller-responsive parent in five, their prevalence is unknown. Next: multi-parent evolution and non-stationary, diversity-driven damage co-evolution — this benchmark is their controlled baseline.

Conclusion. In regenerative NCAs under adversarial recurring damage, robustness comes from channel-aware training (5/5 seeds); evolved modulation is a parent-dependent tonic calibration —

noise-level on most parents, beneficial on the fragile one, parent-locked under transfer. Attribute robustness across model seeds, not only damage seeds.

References

- [1] A. Mordvintsev, E. Randazzo, E. Niklasson, and M. Levin. *Growing Neural Cellular Automata*. Distill 5(2):e23, 2020.
- [2] E. Randazzo, A. Mordvintsev, E. Niklasson, M. Levin, and S. Greydanus. *Self-classifying MNIST Digits*. Distill, 2020.
- [3] E. Niklasson, A. Mordvintsev, E. Randazzo, and M. Levin. *Self-Organising Textures*. Distill 6(2), 2021.
- [4] J. Stovold. *Neural Cellular Automata Can Respond to Signals*. ALIFE 2023; arXiv:2305.12971.
- [5] S. Sudhakaran, E. Najarro, and S. Risi. *Goal-Guided Neural Cellular Automata: Learning to Control Self-Organising Systems*. arXiv:2205.06806, 2022.
- [6] A. Masumori, H. Sato, and T. Ikegami. *Structured Fluctuations and the Information Dynamics of Self-Maintenance in Growing Neural Cellular Automata*. arXiv:2607.12403, 2026.
- [7] J. Lehman and K. O. Stanley. *Abandoning Objectives: Evolution Through the Search for Novelty Alone*. Evolutionary Computation, 19(2):189–223, 2011.
- [8] J.-B. Mouret and J. Clune. *Illuminating Search Spaces by Mapping Elites*. arXiv:1504.04909, 2015.
- [9] R. Wang, J. Lehman, J. Clune, and K. O. Stanley. *Paired Open-Ended Trailblazer (POET): Endlessly Generating Increasingly Complex and Diverse Learning Environments and Their Solutions*. arXiv:1901.01753, 2019.
- [10] M. Dennis, N. Jaques, E. Vinitzky, A. Bayen, S. Russell, A. Critch, and S. Levine. *Emergent Complexity and Zero-Shot Transfer via Unsupervised Environment Design*. NeurIPS 2020.
- [11] J. Parker-Holder, M. Jiang, M. Dennis, M. Samvelyan, J. Foerster, E. Grefenstette, and T. Rocktäschel. *Evolving Curricula with Regret-Based Environment Design*. ICML 2022; arXiv:2203.01302.
- [12] N. Hansen. *The CMA Evolution Strategy: A Comparing Review*. In *Towards a New Evolutionary Computation*, Studies in Fuzziness and Soft Computing vol. 192, pages 75–102. Springer, 2006.
- [13] R. T. Lange. *evosax: JAX-Based Evolution Strategies*. arXiv:2212.04180, 2022.

A Full per-seed, per-condition results

Tonic transplant. Replacing each donor controller by its realized mean m_t injected as a constant (same protocol, both replicas, 3 condition seeds) reproduces the controller’s outcome in 36/40 off-diagonal cells and 10/10 diagonal cells (46/50 overall; Figure 1B) — including every lethal cell, where the transplant kills exactly as the controller does. The four mismatches are magnitude differences inside already-lethal cells. Under the tested regime, the controllers’ small dynamic residual did not measurably contribute, on-parent or off (full CSVs in the repository).

AUC mirrors the same ordering in every seed (full CSVs in the repository).

Table A1: Final Hamming (mean $\pm$ SD, 5 condition seeds $\times$ 8 held-out damage seeds) and survival for each of the ten runs (5 parent seeds $\times$ 2 independent controller evolutions) under hard recurring multi-block damage, $T=2000$. Conditions: $K=3$ parent with modulation pinned to zero; the parent’s own evolved controller; the $K=0$ parent.

Run	$K=3, m=0$		$K=3, \text{own ctrl}$		$K=0$	
	H	surv	H	surv	H	surv
s0.e1	0.0290 $\pm$.0135	1.000	0.0303 $\pm$.0134	1.000	0.0361 $\pm$.0137	1.000
s0.e2	0.0294 $\pm$.0144	1.000	0.0292 $\pm$.0136	1.000	0.0354 $\pm$.0109	1.000
s1.e1	0.0384 $\pm$.0266	0.975	0.0286 $\pm$.0177	1.000	0.1628 $\pm$.0389	0.075
s1.e2	0.0353 $\pm$.0256	0.975	0.0301 $\pm$.0210	1.000	0.1746 $\pm$.0356	0.050
s2.e1	0.0183 $\pm$.0145	1.000	0.0225 $\pm$.0149	1.000	0.0299 $\pm$.0178	1.000
s2.e2	0.0208 $\pm$.0170	1.000	0.0258 $\pm$.0161	1.000	0.0296 $\pm$.0174	1.000
s3.e1	0.0281 $\pm$.0177	1.000	0.0332 $\pm$.0165	1.000	0.0491 $\pm$.0084	1.000
s3.e2	0.0289 $\pm$.0182	1.000	0.0321 $\pm$.0164	1.000	0.0525 $\pm$.0136	1.000
s4.e1	0.0332 $\pm$.0233	1.000	0.0336 $\pm$.0223	1.000	0.0420 $\pm$.0111	1.000
s4.e2	0.0332 $\pm$.0225	1.000	0.0345 $\pm$.0222	1.000	0.0452 $\pm$.0093	1.000

Table A2: Cross-parent transfer probe (unchanged from the per-parent study): the July donor controller evaluated on each recipient parent. Final Hamming and survival on held-out damage seeds; transfer is a penalty in 5/5 recipients, lethal in 2.

Recipient seed	Final Hamming	Survival
0	0.036	1.00
1	0.249	0.00 (lethal)
2	0.434	0.00 (lethal)
3	0.100	0.45
4	0.041	1.00

B Controller-output (m_t) diagnostics

The tonic vector alone does not predict benefit: seeds 1 and 4 emit nearly identical tonic vectors (cosine similarity 0.993, Euclidean distance 0.0065), yet only seed 1 benefits from its controller. The benefit arises from the interaction between a tonic calibration and parent-specific local dynamics, not from a distinctive chemical operating point.

C Pilot: single-parent evaluation hides parent-locking

A three-parent pilot (2026-08-16) first exposed the confound. The single-parent controller from our original five-condition study transferred lethally to two of three sibling parents (survival 0.00, final Hamming 0.249 and 0.434), while zero-output channel parents beat the $K=0$ baseline in 3/3 seeds (0.029/0.034/0.020 vs 0.034/0.177/0.028). The original table’s headline gap was therefore part parent-training effect and part parent-seed luck — motivating this study.

D Model and training details

The grid state $x_t \in [0, 1]^{96 \times 96 \times 16}$ holds 16 channels; the last four are RGBA with alpha last. Perception uses three fixed kernels (identity, Sobel- x , Sobel- y) per channel, giving $p_t \in \mathbb{R}^{48}$:

$$p_t = (K_{\text{id}} * x_t, K_{S_x} * x_t, K_{S_y} * x_t). \quad (1)$$

The update MLP maps $(48+K)$ inputs to 128 hidden units (ReLU) and back to 16 channel increments, with the final layer zero-initialized. Cells update with probability 0.5; a cell is alive when its alpha exceeds 0.1 within its 3×3 max-pooled neighborhood; dead cells are masked to zero.

Table A3: Full 5×5 cross-parent transfer matrix among the five defense parents, both controller replicas (top: e1; bottom: e2). Rows: donor controller; columns: recipient parent. Cells give mean final Hamming / worst-case survival over 3 condition seeds × 8 held-out damage seeds (24 rollouts per cell). Diagonal (bold): recipient’s own controller (survival 1.00 throughout). [†]lethal (survival 0.00). Across the 40 off-diagonal cells: 23 harmful, 15 indistinguishable from zero-output, 2 beneficial — the beneficial cells are exclusively s4→s1, the tonic-aligned pair (cosine +0.99), in both replicas.

e1 donor	recip. s0	recip. s1	recip. s2	recip. s3	recip. s4
s0	0.0315	0.470/0.00 [†]	0.097/0.38	0.028/1.00	0.052/0.88
s1	0.035/1.00	0.0306	0.023/1.00	0.049/1.00	0.035/1.00
s2	0.029/1.00	0.043/0.88	0.0201	0.033/1.00	0.092/0.50
s3	0.035/1.00	0.036/0.88	0.208/0.00 [†]	0.0295	0.044/1.00
s4	0.034/1.00	0.030/1.00	0.023/1.00	0.044/1.00	0.0340
e2 donor	recip. s0	recip. s1	recip. s2	recip. s3	recip. s4
s0	0.0291	0.108/0.25	0.056/0.88	0.032/1.00	0.039/1.00
s1	0.032/1.00	0.0292	0.025/1.00	0.049/1.00	0.032/1.00
s2	0.041/1.00	0.074/0.62	0.0219	0.033/1.00	0.113/0.25
s3	0.028/1.00	0.058/0.75	0.239/0.00 [†]	0.0262	0.028/1.00
s4	0.030/1.00	0.030/1.00	0.020/1.00	0.033/1.00	0.0336

Table A4: Tonic output of the evolved controllers: per-channel mean of m_t , averaged over the two independent evolutions per seed. All ten controllers sit at nonzero, parent-distinct offsets; within-rollout per-channel std of m_t is 0.0002–0.0046 everywhere — flat, tonic output with no lesion-locked response in 10/10 runs.

Seed	mean(m) per channel (avg. of 2 runs)		
0	−0.021	+0.002	+0.027
1	+0.030	+0.012	−0.026
2	−0.026	−0.032	−0.015
3	+0.001	+0.032	−0.010
4	+0.028	+0.007	−0.025

Tonic and phasic modulator dynamics:

$$m_t^{(\text{tonic})} = \alpha m_{t-1}^{(\text{tonic})} + (1 - \alpha) c_t, \quad m_t^{(\text{phasic})} = m_{t-1}^{(\text{phasic})} \cdot e^{-\Delta t / \tau}, \quad (2)$$

with $\alpha = 0.95$, $\tau = 20$, and the injected level the clipped sum, broadcast to all cells and concatenated to perception (input dim $48 + 3 = 51$).

Parents train on a 40×40 emoji target zero-padded to the canvas, with pool-based training (pool 1024, batch 8, worst-in-batch reseeding), random square-erasure damage mid-episode, MSE loss on RGBA, and Adam at 10^{-3} for 8000 steps. A first attempt at 2×10^{-3} diverged to NaN near step 5532 via a late-stage loss-spike cascade; 10^{-3} absorbs the identical spike.

E Perception-radius sweep

A single-lesion sweep (radii {2, 4, 8, 16}, single- and multi-site, with and without modulator channels) maps the boundary of local repair. Recovery is near-perfect for small lesions and degrades once lesions exceed the effective perception radius. Two failure signatures motivate this work: severed fragments that survive but cannot reattach, die, or decide what to grow into, and persistent half-alpha scar tissue at wound sites. Both are information failures: the misbehaving cells are exactly the cells cut off from global context.

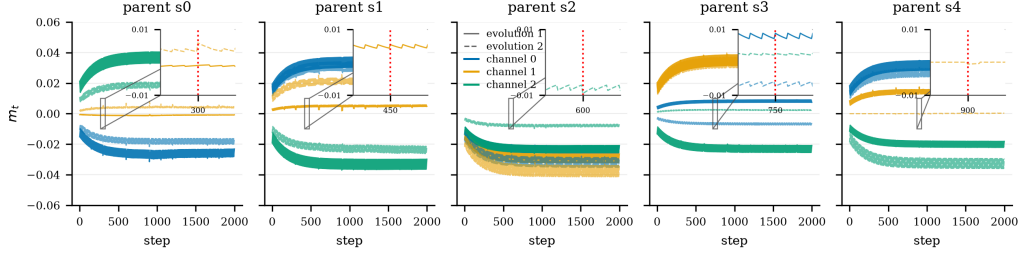


Figure A1: Controller output m_t over a full evaluation rollout, per parent seed, both independent evolutions overlaid (solid/dashed), one line per modulator channel. After a brief initial transient while the perceptual state settles, every controller holds a constant, parent-distinct offset through all lesion events — tonic calibration, not an event-locked policy. Note that the two independent evolutions of the same parent sometimes settle at slightly different setpoints.

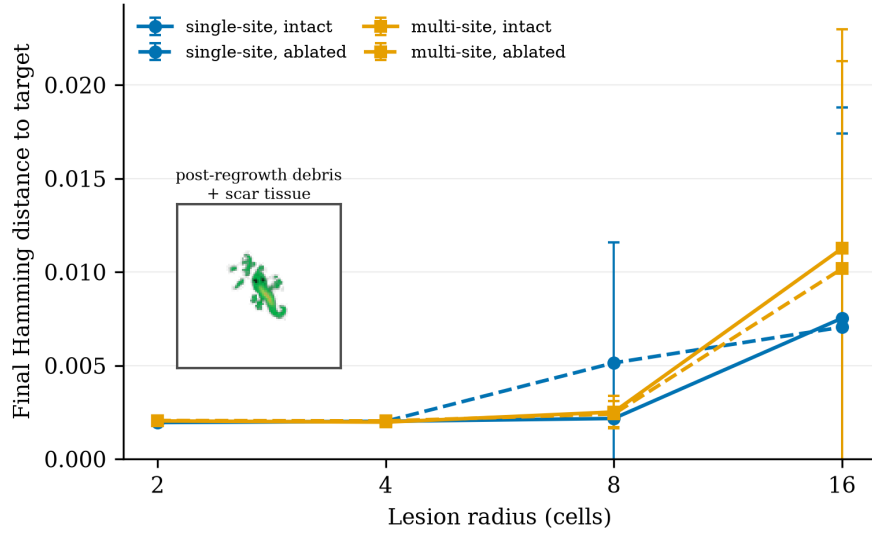


Figure A2: E1 lesion sweep. Final Hamming distance as a function of lesion radius and kind, with and without modulator channels (mean $\pm$ SD over 5 damage seeds). Inset: post-regrowth debris and scar tissue.

F Calibration probes

Collapse probe. The target mask is only 4.6% alive, so a fully-dead grid scores Hamming ≈ 0.046 — close to a struggling organism’s ≈ 0.02 . We verified empirically that the all-dead state scores 0.0461 against struggling-neutral’s 0.0205: death loses by 2.2 $\times$, so the alive-fraction floor is disabled (0.0). Notably, the runbook-default floor of 0.3 would have *penalized every healthy candidate* (a correct lizard lives at ≈ 0.057 alive), pushing evolution toward overgrowth — a case where calibrating from the experiment’s own data overrode a plausible default.

Initial step-size probe. Perturbing the neutral controller by Gaussian noise at increasing σ (25 samples each, hard-regime evaluation) showed: $\sigma \leq 0.01$ stays healthy (some samples beat neutral); $\sigma = 0.05$ is bimodal; $\sigma = 0.3$ lands *every* sample in saturated overgrowth (alive $\rightarrow 0.96$). A first evolution run with the library-default $\sigma_0 = 0.3$ froze at fitness 0.150 from generation 2 onward — 7 $\times$ worse than doing nothing — because CMA-ES was ranking overgrowth-degree noise. $\sigma_0 = 0.01$ brackets the neutral point and the same budget converged smoothly (Figure A3).

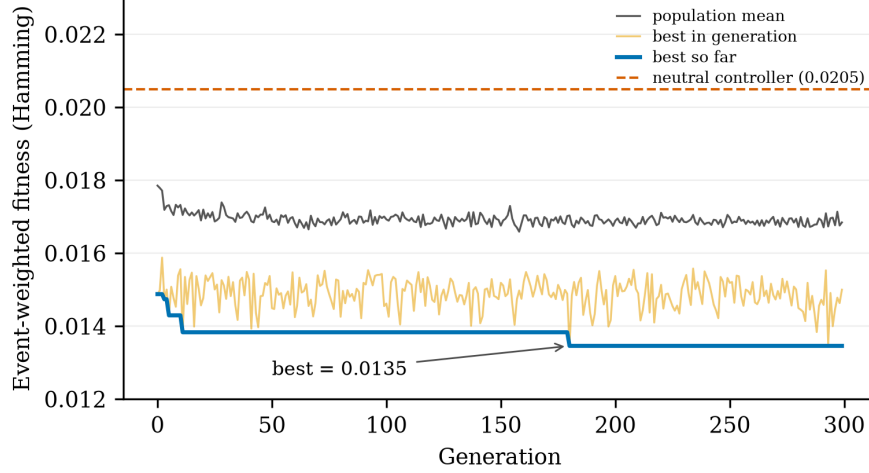


Figure A3: CMA-ES fitness over 300 generations ($\sigma_0=0.01$). Dashed line: neutral controller (0.0205). Best evolved fitness: 0.0135 — a 34% improvement with no stall.

G Evolution objective

We evolve the controller with CMA-ES [12] (Evojax implementation [13]), population 64, initial step size $\sigma_0=0.01$, 300 generations, neutral (all-zero) controller as the initial mean. Fitness is the event-weighted mean Hamming distance to the target alpha mask over full rollouts on the eight train damage seeds:

$$f(\theta) = \frac{1}{\sum_t w_t} \sum_{t=1}^T w_t H_t(\theta), \quad w_t = \exp\left(-\frac{t - \tau_{\text{last}}(t)}{\tau_w}\right), \quad (3)$$

where $\tau_{\text{last}}(t)$ is the most recent lesion step and $\tau_w = 150/3 = 50$. The recency kernel resets at each lesion, so slow repair is expensive even when endpoint Hamming is identical — it de-saturates the metric and prices repair speed directly. Fitness is not the reported metric: all results report unweighted trajectory statistics on held-out seeds. Two calibration controls make this landscape interpretable — a collapse probe (verifying the objective cannot be gamed by alpha suppression) and an initial step-size ablation (showing $\sigma_0=0.3$ saturates the first population in overgrowth where no repair signal exists) — both detailed in Appendix F.

H Single-parent five-condition study

Five conditions share the same protocol, horizon, and damage seeds: **closed-loop** (evolved controller); **static** (the evolved controller evaluated once at $t=0$ and held constant — isolating temporal modulation from the evolved tonic level itself); **constant** (fixed tonic, grid-searched over $\{-1, -0.5, 0, 0.5, 1\}$ on train seeds); **random** (uniform $[-1, 1]$ each step, actuation-matched noise); and **no modulation** ($K=0$ parent). All experiments ran on a single rented A100; parent training took ≈ 20 min/model, evolution ≈ 2.5 h, total project cost under \$15.

Modulation helps substantially (RQ1). Relative to no modulation, modulated conditions recover $2.2\times$ more completely (final Hamming 0.028–0.030 vs. 0.063) and sustain $\approx 4\times$ less cumulative damage (AUC 0.016–0.017 vs. 0.064). The wounds exceed the perception radius, so this improvement is attributable to the broadcast channel — the only non-local pathway — repairing damage that local rules cannot diagnose.

A regime boundary, not a failed ablation (RQ2). The three modulated conditions are statistically indistinguishable: final Hamming spans 0.028–0.030, half-life 6.4–7.2 steps, all differences within noise (pairwise effect sizes < 0.15). We read this as a measurement of the *regime*, not a failure of the controller: the damage process is stationary and memoryless — events are i.i.d. in size, number, and position — so the optimal release policy is time-invariant and a fixed tonic gain is structurally adequate. There is no temporal structure for adaptive scheduling to exploit. Evolution’s measured contribution is automatic discovery of the near-optimal tonic level in 2.5 hours from a neutral start, without the hand search the constant condition required.

Random modulation is lethal. Random actuation kills every run (survival 0.00, final Hamming 0.825): pilot probes show constant levels alone swing mean Hamming from 0.003 to 0.93. The modulator channels are a high-gain control pathway, not a free architectural bonus — the identity of the release schedule determines whether the organism lives.

Metric artifacts of long-horizon evaluation (RQ3). Two artifacts surfaced only under repeated damage. First, repair half-life is defined per lesion relative to each post-lesion jump; on the baseline’s chronically damaged morphology, new lesions often land on already-degraded regions, producing no measurable jump and scoring as zero half-life — so the baseline’s 2.7 steps does *not* indicate fast repair. Second, residual Hamming in all conditions is partly attributable to locomotion drift during regeneration rather than permanent damage; the metric conflates spatial translation with morphological error. Both are documented so that future benchmark users can avoid them.

I Reproducibility

Parents train in ≈ 20 –30 min each on one A100; the defense study ran two independent 300-generation controller evolutions per parent (10 total), each ≈ 2.5 –3 h on the study A100, for a total defense-study cost of $\approx \$25$ of rented GPU time. The transfer matrix is evaluation-only over existing artifacts: 40 off-diagonal evaluations, ≈ 2 minutes of A100 time per full 5×5 replica matrix. Preregistration, analysis script, per-seed artifacts (controllers, parents, trajectories, m_t series), and the mechanical outcome output are in the repository, timestamped before the results. Code will be released upon publication.